

Probiotics, prebiotics infant formula use in preterm or low birth weight infants: a systematic review

Background Previous reviews (2005 to 2009) on preterm infants given probiotics or prebiotics with breast milk or mixed feeds focused on prevention of Necrotizing Enterocolitis, sepsis and diarrhea. This review assessed if probiotics, prebiotics led to improved growth and clinical outcomes in formula fed preterm infants. **Methods** Cochrane methodology was followed using randomized controlled trials (RCTs) which compared preterm formula containing probiotic(s) or prebiotic(s) to conventional preterm formula in preterm infants. The mean difference (MD) and corresponding 95% confidence intervals (CI) were reported for continuous outcomes, risk ratio (RR) and corresponding 95% CI for dichotomous outcomes. Heterogeneity was assessed by visual inspection of forest plots and a chi² test. An I² test assessed inconsistencies across studies. I² > 50% represented substantial heterogeneity. **Results** Four probiotics studies (N=212), 4 prebiotics studies (N=126) were included. **Probiotics:** There were no significant differences in weight gain (MD 1.96, 95% CI: -2.64 to 6.56, 2 studies, n=34) or in maximal enteral feed (MD 35.20, 95% CI: -7.61 to 78.02, 2 studies, n=34), number of stools per day increased significantly in probiotic group (MD 1.60, 95% CI: 1.20 to 2.00, 1 study, n=20). **Prebiotics:** Galacto-oligosaccharide / Fructo-oligosaccharide (GOS/FOS) yielded no significant difference in weight gain (MD 0.04, 95% CI: -2.65 to 2.73, 2 studies, n=50), GOS/FOS yielded no significant differences in length gain (MD 0.01, 95% CI: -0.03 to 0.04, 2 studies, n=50). There were no significant differences in head growth (MD -0.01, 95% CI: -0.02 to 0.00, 2 studies, n=76) or age at full enteral feed (MD -0.79, 95% CI: -2.20 to 0.61, 2 studies, n=86). Stool frequency increased significantly in prebiotic group (MD 0.80, 95% CI: 0.48 to 1.1, 2 studies, n=86). GOS/FOS and FOS yielded higher bifidobacteria counts in prebiotics group (MD 2.10, 95% CI: 0.96 to 3.24, n=27) and (MD 0.48, 95% CI: 0.28 to 0.68, n=56). **Conclusions** There is not enough evidence to state that supplementation with probiotics or prebiotics results in improved growth and clinical outcomes in exclusively formula fed preterm infants.